Human-Computer Interaction

3. Understanding Tasks – Part 2



Hall of Fame or Shame?

Sony Google TV Remote



Hall of Shame!

Sony Google TV Remote



Very complex

So large that it requires two hands

So many controls that you can't use in the dark (watching movie?)

Two navigation pads. When to use which?

But, does have typing input…

Hall of Fame or Shame?

Apple TV Remote



Hall of Shame!

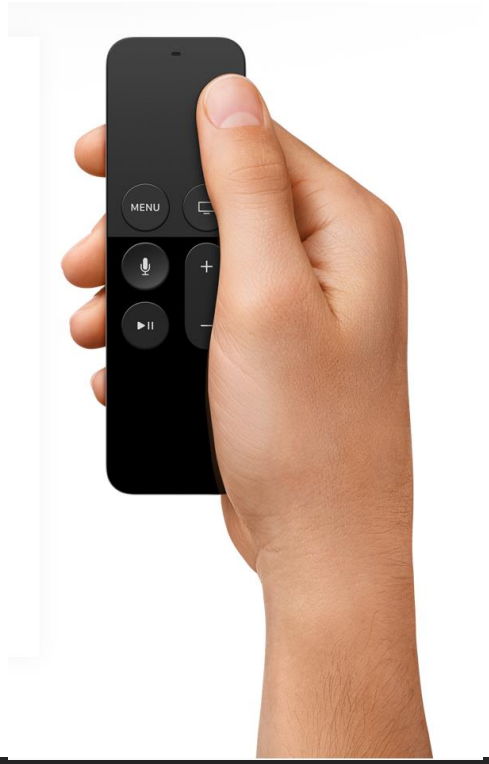
Apple TV Remote



- Overly simple
- Many things require navigating menus
- Text entry is almost impossible
- So small that it is easily lost
- Common tasks easy

Hall of Fame or Shame?

New Apple TV Remote



Still limited number of buttons, but adds

Voice Input

Touch pad w/ navigation, swipes & clicks

Slightly larger → no longer lost in the cushions?

Last week

- Design Thinking Process
- Understanding the user: Who are the users?

Recap: Design Thinking Process



Explore; getting to know your customer

Pinpoint and define the need

Develop a large list of ideas (casting a wide net)

Develop and refine rough ideas

Measure consumer/user reactions to early-stage product concepts, designs, or prototypes

Recap: Who are the users?

Those who interact directly with the product

Those who manage direct users

Those who receive output from the product

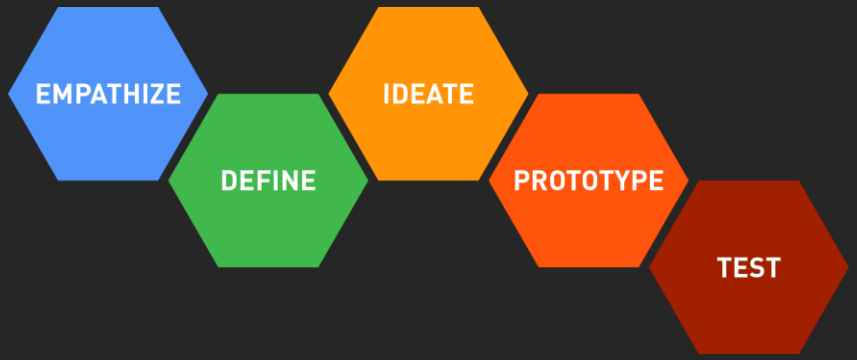
Those who make the purchasing decision

Those who use competitors' product



Stakeholders

- Primary users: people who use the product directly
- Secondary users: people who are affected by the product, or who influence its development



Define:
Understanding Task

Task Analysis

- *Important note:* Only a system a user can use will be sold
- → What are the user group's **tasks** to be solved with the system?

- A **task** is a meaningful unit during a user's activity; they lead to a meaningful (partial) results

- Quasi-Synonyms:
  Task, Scenario, Use-Case



Task Analysis

- "Define" the needs
- The process of examining the way in which people perform their tasks
- Deliverable（可交付的）: a diagram explaining the steps that a user (must) takes in order to complete a goal
  - You can depict the actions taken by the users to help them achieve their goals
  - You can discover what the system will be used for and what functionalities the system must provide

→ **You can identify where additional user support is required or eliminate unnecessary steps**

Task Analysis

Task analysis is useful for understanding:

- Users' goals and what they are trying to achieve
- The steps that your users currently take in order to achieve their goals
- The personal, social and cultural experiences that users bring to the tasks
- The influence of the physical environment on the users while attempting to meet a goal

Task analysis

A sequence diagram is a type of interaction diagram because it describes how—and in what order—a group of objects works together.

UML: Unified modeling language

- Various ways of documenting tasks:
  - Table: **User groups** X **tasks**
    for each note down **frequency** of the task
  - Table: Objects X tasks
  - For complex tasks, e.g., sequence diagrams (UML)
  - Describe a task according to a given schema (not mandatory)





https://www.geeksforgeeks.org/unified-modeling-language-uml-sequence-diagrams/

Task analysis – Example 1

Table with user groups and tasks, as well as the relative frequency of each task

User **Task**	Typist	Secretary	Administrator	…
Create letter	often	often	occasionally	
Create report	never	occasionally	often	
Insert figure	never	occasionally	often	
print	often	often	often	
…				

Assign objects to tasks – Example 1

	Bill	Letter	Article	…
Create	+	+	+	
Print	+	+	+	
Insert table	-	-	+	
Insert figure	-	-	+	
Spellcheck	-	+	+	
Reminder	+	-	-	
…				

Types of tasks

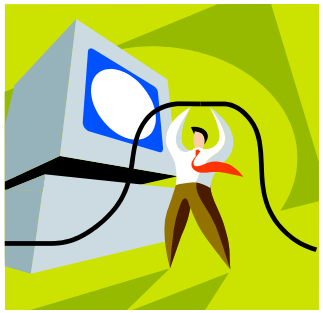
- Specific types of tasks
  - Collaborative tasks
    - in sequential order
    - In parallel (cooperative)
    - Can be described with flow diagrams
  - Unstructured and explorative tasks:
    - Do not restrict the range of operations too much! Allow user to “experiment”!
  - Content-driven tasks
    - Multimedia, games, etc.



Task profiles



- Goal: Find out what the user wants to do!
  - User needs, goals, an resulting tasks
- User think in ‘tasks’, not in ‘functions’!
- Example
  - Task: I want to change my address
  - Function:
    - First, I need to log in,
    - then I need to navigate to settings,
    - select the item “change address”,
    - change address,
    - and then click on “save”



Focus on Task Analysis

When collecting data for task analysis, focus on:

- Trigger: What prompts users to start their task?
- Desired Outcome: How users will know when the task is complete?
- Base Knowledge: What will the users be expected to know when starting the task?
- Required Knowledge: What the users actually need to know in order to complete the task?
- Artifacts: What tools or information do the users utilize during the course of the task?

Task Analysis: Steps

1. Identify the task to be analyzed: What is that user's goal and motivation for achieving it?
2. Break this goal (high-level task) down into subtasks: You should have around 4–8 subtasks after this process. If you have more, then it means that your identified goal is too high-level and possibly too abstract
3. Collect data to identify tasks/subtasks that the user performs
4. Draw a layered task diagram of each subtask and ensure it is complete.

Task analysis: Questions

1. Who is going to use system?
2. What tasks do they now perform?
3. What tasks are desired?
4. How are the tasks learned?
5. Where are the tasks performed?
6. What's the relationship between user & data?
7. What other tools does the user have?
8. How do users communicate with each other?
9. How often are the tasks performed?
10. What are the time constraints on the tasks?
11. What happens when things go wrong?

Task Analysis: Example

Goal: 1. Send a text message to hospital's system

Tasks/Subtasks:

1.1. Open the messaging application on her mobile phone.

1.2. Enter the hospital system's special number.

1.3. Move to the text input field.

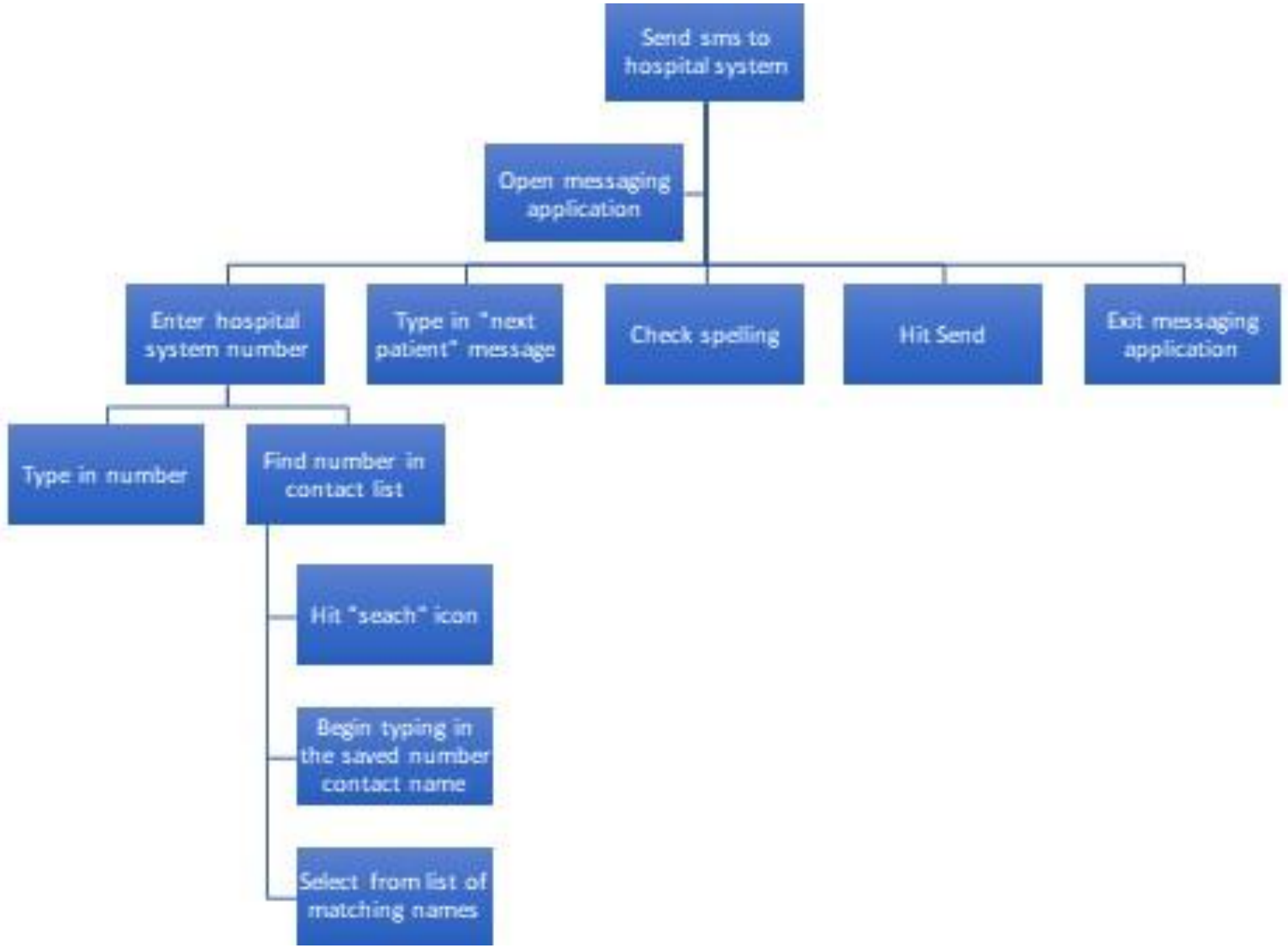
1.4. Type the words "next patient".

1.5. Check the spelling (because it needs to be precise for the system to accept it).

1.6. Hit the "send" button.

1.7. Exit the messaging application.

Task Analysis: Example



Task profiles – guidelines

- Tasks that need to be supported should be determined starting the design profiles
- (Unnecessary) tasks should not be added just because they are easy to implement
- Task frequency (according to the user profiles) should result in design decisions
- The more frequent a function is used, the easier it should be to access.

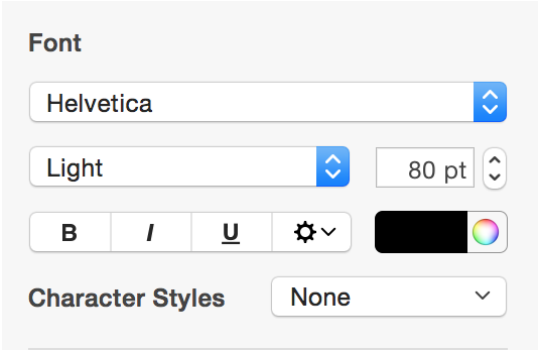


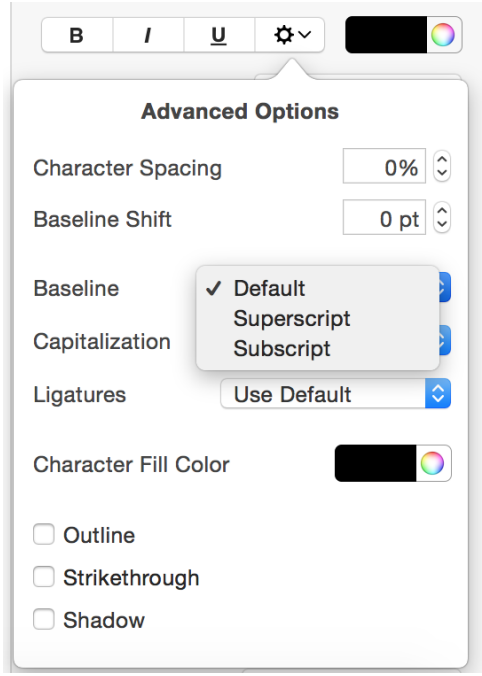
Quelle: http://mitschau.edu.lmu.de/video_online/vorlesungen/wise2006_2007/mensch_maschine/index.php

Task frequency

- **Bold** can be found directly in the sidebar
- **Subscript** needs to extra clicks
- The assumption is that users need more often **bold** than **subscript**
- (Customization and/or shortcuts can be used for users with specific demands)

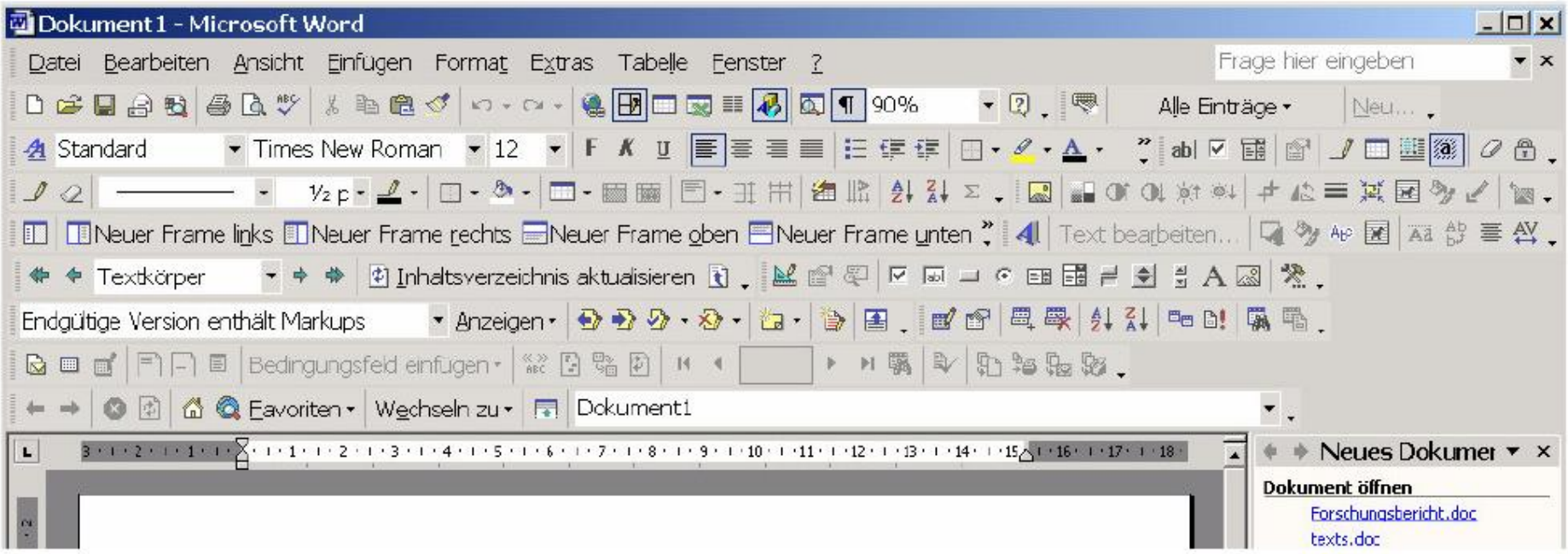
Apple Keynote





Task frequency – discussion!

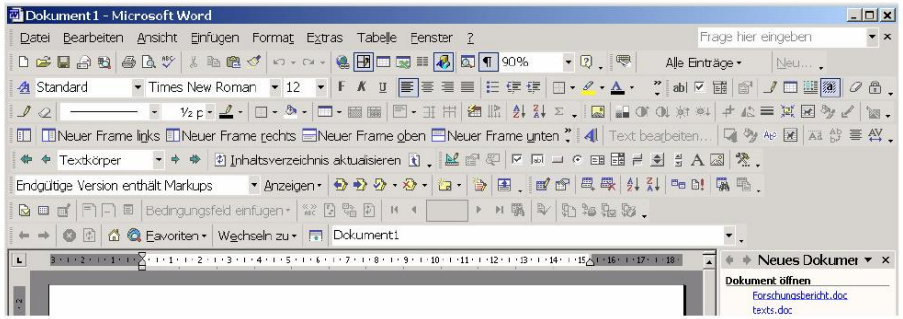
- Tradeoff between fast access and cluttered/crammed design
- Example toolbar:



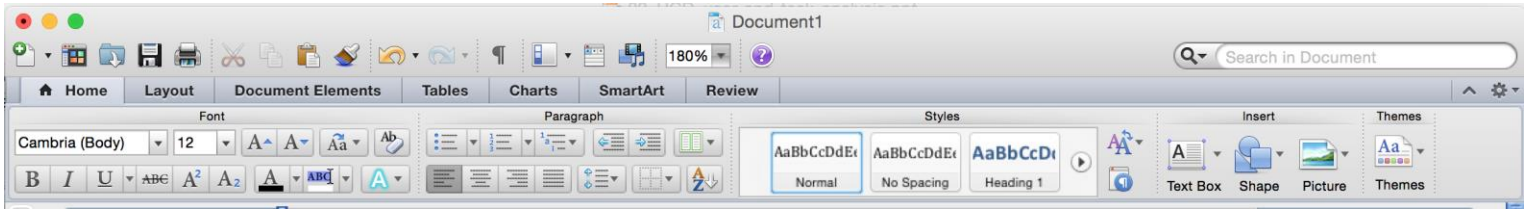
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Task frequency

- Tradeoff between fast access and cluttered/crammed design
- Example toolbar:
  - The more functions are directly visible, the faster the access
  - However, the time to find them gets longer (at least initially)
  - More screen space is used up
    (trend: larger displays OR smaller displays)



Word 2000 (or so)



Word 2011 for Mac

Quelle: http://mitschau.edu.lmu.de/video_online/vorlesungen/wise2006_2007/mensch_maschine/index.php

Task description (1)
Schema for task specification

- Pre-conditions for the task
  - *Example:* state of objects: to revise a document it must already exist
- Goals of the task
  - Can be expressed via status of objects
  - *Example:* generate a letter with the content that was just dictated to the secretary
- "Breakdowns" in current workflows
  - What are problematic steps in the current system
  - Room for improvement in the new system
- Post-condition of the task
  - States of objects that were involved in the task
  - *Example:* after generating a letter, the author becomes the file owner

Task description (2)
Schema for task specification

- Definition of actors/agents
  - Classes of people that are involved in the task
  - Agents describe roles of users
  - *Example:* typist, manager
- Definition of objects
  - In the current and envisioned system
- Steps in task execution
  - Task decomposition in its elementary components
- Critical characteristics of the task
  - Important steps, even though if they might get automated in the new system

Task analysis: NJ Transit ticket machine

Goals:

- Buy new ticket
- Add value to ticket
- Pay with: Debit, Credit, Cash



Task analysis: Questions

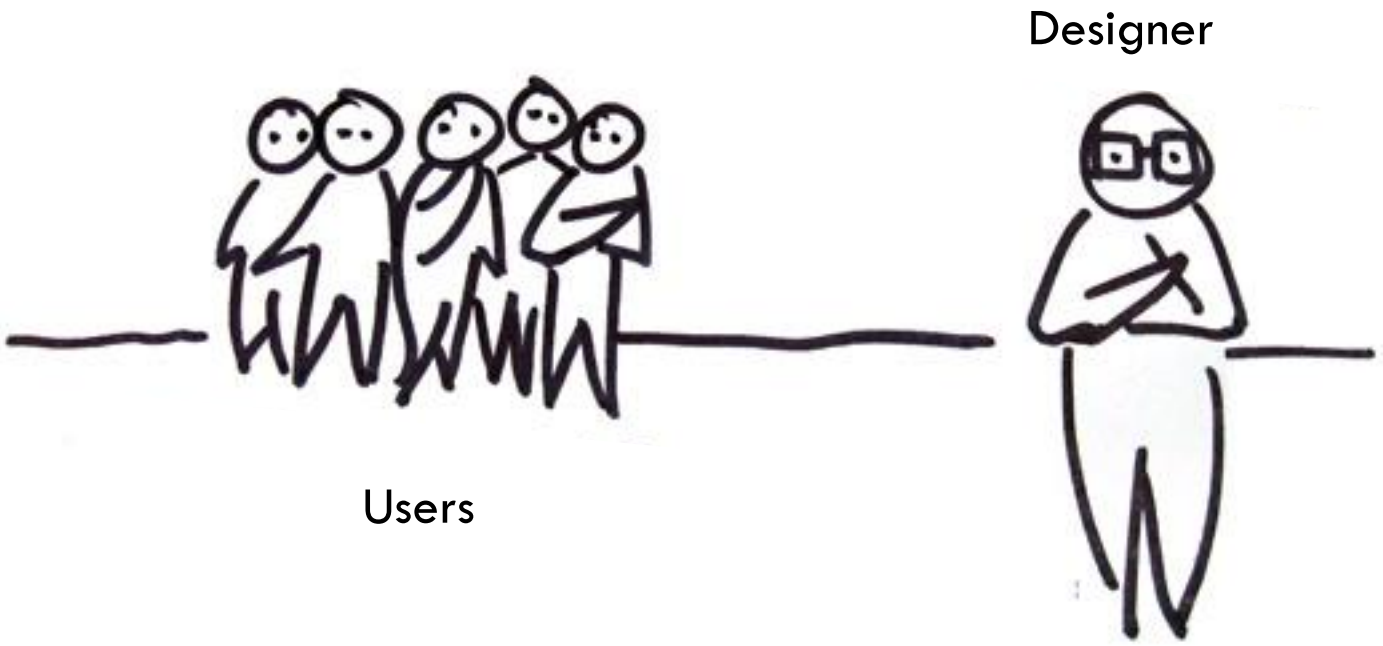
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Common Errors in Task Analysis

- Thinking from the system's point of view, rather than the user's
  - "Notify user about appointment"
  - vs. "Get a notification about appointment"
- Fixating too early on a UI design vision
  - "The system bell will ring to notify the user about an appointment..."
- Bogging down in *what* users do now (**concrete** tasks), rather than *why* they do it (**essential** tasks)
  - "Save file to disk"
  - vs. "Make sure my work is kept"
- Duplicating a bad existing procedure in software
- Failing to capture good aspects of existing procedure

How can we understand the users and their needs?



We (designers) are not the users!

Class Exercise

Conduct a task analysis:

选课

Group project

P1. Problem Statement

- Identify a real problem from the given themes that a smartphone app can solve. To do this, your group needs to brainstorm a bit.
- Focus on the problem. You don't know the solution yet!
- Describe the problem you are trying to solve with a new smartphone application. Your description should include:
  1. What is the context?
  2. What is the problem?
  3. Why is it an important problem to solve?
  4. Why is a smartphone app a proper solution (e.g., why not a website)?

- Length: You will need to fill up at least two paragraphs to fully explain the contents above. Less than one page
- Submission: Turn in your proposal to Canvas
- Due: midnight tomorrow (9/25) – You will have a revision cycle if needed

P1. Problem Statement

Grading Rubric

- Novelty of an idea (2pt)
- Feasibility of completion (1.5pt)
- Within the scope of given topics (1.5pt)

Late Submission Penalty

- -30% within 24 hours after a submission due
- Not accepting late submissions after 24 hours from a submission due

Other Policies

- Do not expect my response to an inquiry over the weekend: Ask peers
- Further instruction can be given verbally during class

Brainstorm Ideas

"Mobile app for health, wellness, and environmental sustainability"

1. Think about ways to promote an environmentally sustainable daily life, such as traffic congestion, energy shortages, and air & water pollution
2. Think about ways to support personal healthcare
3. Think about problems that marginalized populations encounter and ways to support their needs for the quality of life using a smartphone application
4. Think about solutions to facilitate collective efforts in solving important societal problems: Citizen Science

- Consider adding a smart-watch app coupled with a mobile app
- No meetup, buddy-finding apps
- No activity/calorie tracking apps
- Consider accessibility to potential users!